Moving Parameters Optimization-Based Model Predictive Adaptive Impedance Control for Robotic Motion/Force Tracking

Ziling Wang, Lai Zou, Xiaojie Su, and Jiantao Li

***Abstract*—In the robotic grinding system for the turbine blade, the low motion accuracy of robots and the complex contact environment may lead to errors in the grinding force and trajectory, thus reducing the machining quality of the blade. To solve the problem, a model predictive adaptive impedance control method based on moving parameters optimization is developed to improve the robot end's force/motion accuracy. Specifically, a novel adaptive impedance model is first developed and the adaptive parameter involved in the model is determined by stability proving. Then, the impedance model and the robotic kinematic model are, respectively, used in the force prediction and robot end motion prediction to construct a model predictive control (MPC) scheme. Furthermore, we propose a real-time moving parameters optimization method to enhance the force prediction accuracy by adjusting the key parameters in the impedance model, thus improving the force control accuracy of the whole control scheme. Simulation and experiments for the turbine blade in the robotic grinding system are conducted to verify the superiority of the proposed control method. The interaction force accuracy with the proposed method is 0.744 N, and the position accuracy is about 0.447 mm.**

***Index Terms*—Adaptive impedance model, force/motion control, model predictive model, moving parameters optimization, robotic grinding system, turbine blade.**

I. Introduction

ROBOTIC grinding technology is becoming the predominant processing method for aeroengine blades with complex curvature changes, such as turbine blades, due to their high flexibility and integration advantages [1]. However, the low motion accuracy of industrial robots frequently leads to deviations in the grinding trajectory of turbine blades, resulting in significant fluctuations in the grinding force, which ultimately decreases machining quality.

Some researchers have conducted extensive research in the field of robot force/position control. Wang et al. [2] proposed a novel hybrid force/motion controller in the Cartesian space of a robotic manipulator based on advanced fuzzy control to enhance the tracking accuracy of force/motion in the robotic workspace. Kim et al. [3] presented an intrinsic force sensing method for robots, which can acquire external force signals during human–robot interaction and enhance contact force sensing performance. Sun et al. [4] presented an admittance control method to achieve human–robot collaborative (HRC) assembling of large components based on the dynamic model of the used parallel robot. Han et al. [5] designed a modified dynamic movement primitive method for robot control. Shahriari et al. [6] proposed a modular uni-arm control law using a novel adaptation policy, which integrates the intrinsic limitations of some conventional control methods. Sharifi et al. [7] presented a new learning-based time-varying impedance controller to facilitate an autonomous physical human–robot interaction by using the proposed adaptation laws and update rules. Zahedi et al. [8] designed a user-adaptive variable damping controller that enhances the overall performance of coupled human–robot systems. Xu et al. [9] presented a dynamic neural network-based adaptive impedance control method for redundant robots under multiple physical constraints. Yang et al. [10] designed a novel spatial repetitive impedance learning control strategy, which can enhance the performance of physical interactions. The current adaptive impedance control methods for robot end contact force mainly use adaptive models including online neural networks or errors to estimate and compensate for control errors. As a prevalent control strategy, model predictive control (MPC) is increasingly employed in the motion control of industrial robots due to its enhanced performance in trajectory tracking and force control [11], [12].

Some scholars have utilized MPC methods to achieve high-precision trajectory tracking and force control during the robot's movement. Zhu et al. [13] investigated the learning-based MPC problem for nonlinear systems with environmental uncertainties and constraints. Woolfrey et al. [14] proposed a predictive control method for manipulators operating on mobile platforms in Cartesian space.

To achieve force tracking of the robot end in the contact workspace using the MPC, it is essential to segment the contact workspace into force and position subspaces [15]. Anand et al. [16] introduced a novel variable impedance controller based on

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deep model predictive theory for compliant robotic manipulation. Maciej et al. [17] provided a MPC method to allow compliant behavior during the interaction process between the robot end and the environment. Wahrburg and Listmann [18] developed an admittance tracking scheme with consideration of MPC to handle interaction forces between the robot and the environment. Minniti et al. [19] combined an MPC-based controller with two adaptive schemes, derived from online system identification and adaptive control. Gold et al. [20] presented a model predictive interaction control method for robot manipulation tasks constructed based on the repetitive solution of an optimal control problem. Kazim et al. [21] combined a nonlinear model predictive path following admittance control, which describes the desired dynamic interaction between two objects. Zhang et al. [22] designed an online moving horizon-based observer for the interaction model between the robot and the contact environment. Matschek et al. [23] developed a learning-based model predictive force and motion control method using Gaussian processes.

The dynamic equations of the contact force proposed in Anand et al. [16], Maciej et al. [17], and Wahrburg and Listmann [18] do not involve the force errors, which cannot achieve closed-loop force control. However, a stiffness model or damping-stiffness model proposed in Gold et al. [20] and Zhang et al. [22] was used as the dynamic equation, which may cause severe oscillations or steady-state errors in the force control process. In addition, the dynamic equations with fixed impedance parameters proposed in Kleff et al. [15], Maciej et al. [17], Wahrburg and Listmann [18], Gold et al. [20], and Kazim et al. [21] cannot improve the force control robustness of MPC methods in the complex contact environment. However, online optimization methods proposed in Anand et al. [16] and Janine et al. [23] cannot consider the predictive errors of state variables in the MPC schemes, which can limit the prediction accuracy and further reduce the control performance of controllers.

To overcome the above problems, this article develops a novel model predictive adaptive impedance controller based on a moving parameter optimization (MPAIC-MPO) method. The contributions of the proposed controller are illustrated as follows.

1) An impedance model with an adaptive variable is used to construct the dynamic equation of contact force. The adaptive variable in the dynamic can effectively improve the force control robustness and reduce the steady-state errors of the controller by adaptively adjusting the variable based on control errors.
2) A novel MPO method is proposed to online adjust the crucial parameters involved in the force dynamic equation based on predictive errors between the predictive and actual force in historical control cycles. The method can accurately obtain the optimized parameters in the equation using the developed error model, and improve the predictive accuracy of contact force.

II. Seven-Degree of Freedom (DOF) Robotic Grinding System

The symbols used throughout this article are summarized in Table I. Fig. 1 shows the seven-DOF robotic grinding system.

TABLE I
DEFINITIONS OF VARIABLES

Definition	Symbol	Definition	Symbol
Cutter contact points	C	Actual environmental stiffness	k_e
Normal vector	N	Adaptive parameter	ρ
Ideal normal force	f_{Nd}	Control cycle	T
Robot end pose	P	Damping coefficient	b_d
Actual normal force	f_N	Inertial coefficient	m_d
Ideal robot end pose	p_d	Stiffness coefficient	k_d
Actual force vector	$F_N^R \in \mathbb{R}6^{\times 1}$	Force state variable	$\tilde{F}_N^R \in \mathbb{R}^{6\times 1}$
Robotic joint position	$q \in \mathbb{R}^{7\times 1}$	State vector	$v \in \mathbb{R}^{26\times 1}$
Input vector	$u \in \mathbb{R}^{26\times 1}$	Ideal state vector	$v_d \in \mathbb{R}^{26\times 1}$
Weight matrix	$Q \in \mathbb{R}^{26\times 26}$	Weight matrix of the interaction force	$Q_f \in \mathbb{R}^{6\times 6}$
Weight matrix of the robot end pose	$Q_p \in \mathbb{R}^{6\times 6}$	Adaptive rate	σ
Prediction horizon length	N	Moving horizon length	N_b
Input weights	$R \in \mathbb{R}^{6\times 6}$	Force control accuracy	e_F
Position control accuracy	e_P	Posture control accuracy	e_O
Inertial coefficient matrix	$M_d \in \mathbb{R}^{6\times 6}$	Stiffness coefficient matrix	$K_e \in \mathbb{R}^{6\times 6}$
Damping coefficient matrix	$B_d \in \mathbb{R}^{6\times 6}$	Cost function	Ω

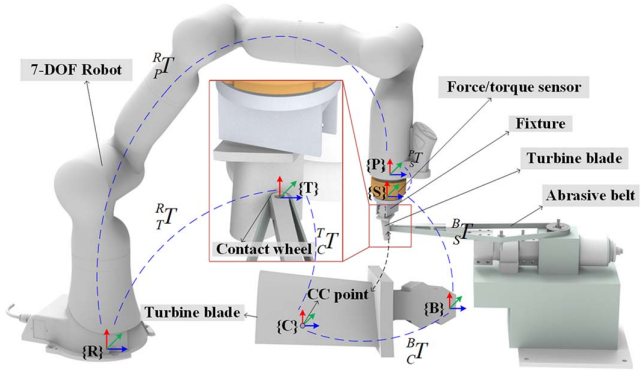


Fig. 1. Robotic grinding platform.

The turbine blade is clamped at the robotic end, while the grinding tool is positioned externally to the robot. The coordinate system distribution of the system is shown in Fig. 1. The workpiece (i.e., the blade) coordinate system is $\{B\}$, the tool coordinate system is $\{T\}$, the cutter contact (CC) point coordinate system is $\{C\}$, the robot base coordinate system is $\{R\}$, the robot tool center point (TCP) coordinate system is $\{P\}$, and $\{S\}$ represents the force transducer coordinate system. The kinematic chain of the grinding system is as follows:

$$ {}^R_P T = {}^R_T T \cdot {}^T_C T \cdot {}^C_B T \cdot {}^B_S T \cdot {}^S_P T \tag{1} $$

where ${}^B_P T \in \mathbb{R}^{4\times 4}$ is the transformation matrix corresponding to the coordinate system $\{P\}$ relative to the coordinate system $\{B\}$.

Fig. 2 illustrates the grinding trajectory of the turbine blade. CC points $\{c_i\}$ $(i = 1, 2, \ldots, n)$ are obtained by discretizing the planned grinding trajectory, as shown in Fig. 2. The chord height errors corresponding to adjacent CC points on the

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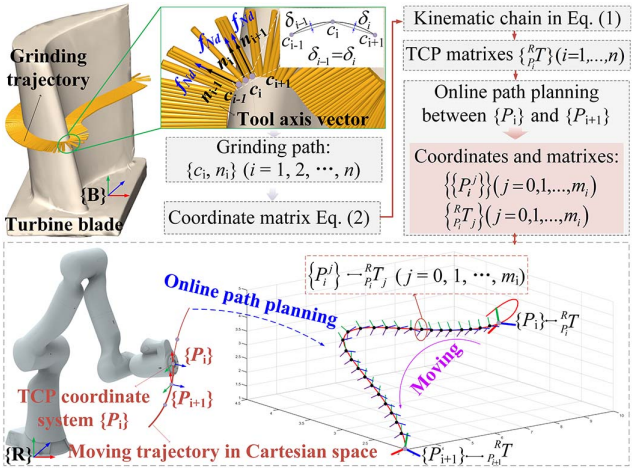


Fig. 2. Trajectory planning for the robotic end in cartesian space.

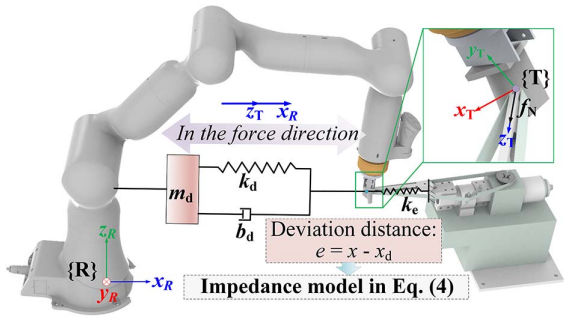


Fig. 3. Impedance model between the robot end and grinding tool.

trajectory are equal, i.e., $\delta_{i-1} = \delta_i$, where δ_i is the chord height error between the CC points c_i and c_{i+1} [1]. Next, the normal vector n_i corresponding to the point c_i is calculated based on the surface information of the blade, meanwhile, the normal vector coincides with the grinding tool axis. The reference information in the control process is $\{c_i, n_i, f_{Nd}\}$ $(i = 1, 2, \ldots, n)$, where f_{Nd} is the expected normal force between the grinding tool and the blade along the normal vector n_i. Let the coordinate system corresponding to CC point c_i be $\{C_i\}$, and the transformation matrix ${}^{B}_{C_i}T$ is calculated by the following equation:

$$ {}^{B}_{C_i}T = \begin{bmatrix} \bar{u}_i & \bar{v}_i & \bar{n}_i & c_i \\ \multicolumn{3}{c}{0} & 1 \end{bmatrix} \tag{2} $$

where $\bar{n}_i \in \mathbb{R}^{3\times 1}$ is the unit vector of n_i. $\bar{v}_i \in \mathbb{R}^{3\times 1}$ is the unit vector of v_i, and $v_i = n_i \times \overrightarrow{c_i c_{i+1}}$. The vector $\bar{u}_i = \bar{n}_i \times \bar{v}_i$. The operation matrix ${}^{B}_{P_i}T$ of the TCP pose corresponding to ${}^{B}_{C_i}T$ is calculated based on (1) as shown in Fig. 2, where $\{P_i\}$ is the TCP coordinate system of the CC point coordinate $\{C_i\}$.

The robot control system is regarded as a discrete system with a control cycle T. The robotic moving trajectory $\{\{P_i^j\}\}$ $(j = 0, 1, 2, \ldots, m_i)$ between the coordinates $\{P_i\}$ and $\{P_{i+1}\}$ is obtained using the online trajectory planning method [24] as shown in the figure, where $\{P_i^0\} = \{P_i\}$, $\{P_i^{m_i}\} = \{P_{i+1}\}$. The motion time between the coordinate systems $\{P_i^j\}$ and $\{P_i^{j+1}\}$ is one control cycle T, and $\{P_i^j\}$ is regarded as the TCP coordinate at jT time. The transformation matrix ${}^{R}_{P_i}T_j \in \mathbb{R}^{4\times 4}$ of $\{P_i^j\}$ relative to coordinate $\{R\}$ is as follows:

$$ {}^{R}_{P_i}T_j = \begin{bmatrix} {}^{R}_{P_i}R_j & P_i^j \\ 0 & 1 \end{bmatrix}. \tag{3} $$

Therefore, the whole ideal trajectory of the robot end in workspace is $L = \left\{\left\{{}^{R}_{P_1}T_j\right\}, \left\{{}^{R}_{P_2}T_j\right\}, \ldots, \left\{{}^{R}_{P_n}T_j\right\}\right\}$ for all CC points $\{c_i\}$ $(i = 1, 2, \ldots, n)$, and the expected TCP pose in every control cycle is obtained based on the trajectory L.

III. MPAIC-MPO Controller

A. Impedance Model

The impedance model between the blade clamped at the robotic end and the grinding tool is established in the z-axis of the tool coordinate system $\{T\}$, which coincides with the x-direction of the base coordinate $\{R\}$, as shown in Fig. 3. The movement DOF along the x-axis is defined as the force subspace, and the movement DOF along the y- and z-axes directions and rotation DOF around the x-, y-, and z-axes are defined as the motion subspace. Therefore, the impedance model in the force subspace is as follows:

$$ f_N - f_{Nd} = m_d(\ddot{x} - \ddot{x}_d) + b_d(\dot{x} - \dot{x}_d) + k_d(x - x_d) \tag{4} $$

where f_N is the actual normal interaction force in the z-direction of the coordinate $\{T\}$. The value f_N is limited by the maximum load of the robot end. The robot end velocity in Cartesian space cannot be too large, and it is less than about 15 mm/s in the experimental system of the research work. The parameter $p \in \mathbb{R}^{6\times 1}$ represents the robot pose and $p = [x, y, z, w, \beta, r]^T$, where w, β, and r are Euler angles to represent the orientation of the robot end in Cartesian space, and x, y, and z are the robot end position. The parameter $p_d \in \mathbb{R}^{6\times 1}$ represents the ideal pose and $p_d = [x_d, y_d, z_d, w_d, \beta_d, r_d]^T$, which is calculated by the ideal trajectory L. The parameters m_d, b_d, and k_d are, respectively, the inertial parameter, damping coefficient, and stiffness coefficient. The interaction model of the external contact environment in the force direction is as follows:

$$ f_N = k_e(x_r - x) \tag{5} $$

where x_r represents the equilibrium position of the robotic end in the x-direction of the coordinate $\{R\}$ and k_e is the actual environmental stiffness. When $k_d = 0$, it can be guaranteed that the steady-state $f_N = f_{Nd}$ can be reached under any contact environment stiffness [25]. Therefore, the impedance model is rewritten as follows:

$$ f_N - f_{Nd} = m_d \cdot \ddot{e} + b_d \cdot \dot{e} \tag{6} $$

where $e = x - x_d$. The adaptive parameter ρ is introduced in (6) and the equation is rewritten as follows:

$$ f_N - f_{Nd} = m_d \cdot \ddot{e} + b_d \cdot (\dot{e} + \rho) \tag{7} $$

where ρ is expressed as follows in the discrete-time domain:

$$ \rho((k+1)T) = \rho(kT) + \sigma \frac{(-f_N(kT) + f_{Nd}(kT))}{b_d} \tag{8} $$

where $\rho(kT)$ is the adaptive parameter in the kth control cycle, and the symbol σ is the adaptive rate. Based on this equation,

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the iterative process for the parameter is expressed as follows:

$$\begin{aligned} &b_d\cdot\rho(T)=b_d\cdot\rho_0+\sigma\cdot(-f_{N0}+f_{Nd0})\\ &b_d\cdot\rho(2T)=b_d\cdot\rho(T)+\sigma\cdot(-f_N(T)+f_{Nd}(T))\\ &\qquad\vdots\\ &b_d\cdot\rho(kT)=b_d\cdot\rho((k-1)T)+\sigma\cdot(-f_N((k-1)T)+f_{Nd}((k-1)T))\\ &\qquad\vdots \end{aligned} \tag{9}$$

where ρ_0 is the initial value of the adaptive parameter, and f_{N0} and f_{Nd0} are, respectively, the actual and expected normal contact force in the initial control cycle. Then the mathematical relationship between the value $b_d\cdot\rho(kT)$ in the kth control cycle and the initial equation $b_d\cdot\rho_0$ is established by the following equation:

$$\begin{aligned} b_d\cdot\rho(kT)=b_d\cdot\rho_0+\sigma\cdot(-f_N((k-1)T)+f_{Nd}((k-1)T))+\cdots\\ +\sigma\cdot(-f_{N0}+f_{Nd0}). \end{aligned} \tag{10}$$

Substituting (5) and (8) into (7)

$$\begin{aligned} &k_e(x_r(kT)-x(kT))-f_{Nd}=m_d\cdot\ddot{e}(kT)+b_d\cdot\dot{e}(kT)\\ &\quad+b_d\cdot\rho((k-1)T)+\sigma\cdot k_e\cdot(x_r((k-1)T)-x((k-1)T))\\ &\quad-\sigma\cdot f_{Nd}((k-1)T). \end{aligned} \tag{11}$$

Differentiating the environmental stiffness model

$$\begin{cases} x=x_r-\dfrac{f_N}{k_e},\ \dot{x}=\dot{x}_r-\dfrac{\dot{f}_N}{k_e},\ \ddot{x}=\ddot{x}_r-\dfrac{\ddot{f}_N}{k_e}\\ e=x_r-\dfrac{f_N}{k_e}-x_d,\ \dot{e}=\dot{x}_r-\dfrac{\dot{f}_N}{k_e}-\dot{x}_d,\ \ddot{e}=\ddot{x}_r-\dfrac{\ddot{f}_N}{k_e}-\ddot{x}_d. \end{cases} \tag{12}$$

Substituting (12) into (11)

$$\begin{aligned} &m_dk_e\cdot(\ddot{x}_r(kT)-\ddot{x}_d(kT))+b_dk_e\cdot(\dot{x}_r(kT)-\dot{x}_d(kT))\\ &\quad+m_d\cdot\ddot{\varepsilon}(kT)+k_e\cdot\varepsilon(kT)+b_d\cdot\dot{\varepsilon}(kT)+k_eb_d\cdot\rho((k-1)T)\\ &\quad+\sigma k_e\cdot\varepsilon((k-1)T)=m_d\cdot\ddot{f}_{Nd}(kT)+b_d\cdot\dot{f}_{Nd}(kT) \end{aligned} \tag{13}$$

where $\varepsilon(kT)=f_{Nd}(kT)-f_N(kT)$ and $\dot{x}_d(kT)=(x_d((k+1)T)-x_d((k)T))/T$. Let $k_e(x_r(kT)-x_d(kT))=f'_{Nd}(kT)$, then (13) is transformed into

$$\begin{aligned} &m_d\cdot\ddot{\varepsilon}(kT)+k_e\cdot\varepsilon(kT)+b_d\cdot\dot{\varepsilon}(kT)+k_eb_d\cdot\rho((k-1)T)\\ &\quad+\sigma k_e\cdot\varepsilon((k-1)T)=m_d\cdot\underbrace{[\ddot{f}_{Nd}(kT)-\ddot{f}_{Nd}'(kT)]}_{\ddot{\delta}(kT)}\\ &\quad+b_d\cdot\underbrace{[\dot{f}_{Nd}(kT)-\dot{f}_{Nd}'(kT)]}_{\dot{\delta}(kT)}. \end{aligned} \tag{14}$$

Substituting (10) into (14)

$$\begin{aligned} &m_d\cdot\ddot{\varepsilon}(kT)+b_d\cdot\dot{\varepsilon}(kT)+k_e\cdot\varepsilon(kT)+\sigma k_e\cdot\begin{pmatrix}b_d\cdot\rho_0+\varepsilon((k-1)T)\\+\varepsilon((k-2)T)+\cdots+\varepsilon_0\end{pmatrix}\\ &=m_d\cdot\ddot{\delta}(kT)+b_d\cdot\dot{\delta}(kT) \end{aligned} \tag{15}$$

where $\varepsilon_0=f_{Nd0}-f_{N0}$. Let the current control cycle kT be t, so the above equation is rewritten as follows:

$$\begin{aligned} &m_d\cdot\ddot{\varepsilon}(t)+b_d\cdot\dot{\varepsilon}(t)+k_e\cdot\varepsilon(t)+\sigma k_e\cdot\begin{pmatrix}b_d\rho_0+\varepsilon(t-T)+\varepsilon(t-2T)\\+\cdots+\varepsilon(t-kT)\end{pmatrix}\\ &=m_d\cdot\ddot{\delta}(t)+b_d\cdot\dot{\delta}(t) \end{aligned} \tag{16}$$

where $\varepsilon(t-kT)=\varepsilon_0$. Performing a Laplace transform on the equation

$$\begin{aligned} &m_d\cdot\varepsilon(s)s^2+b_d\cdot\varepsilon(s)s+k_e\cdot\varepsilon(s)\\ &+\sigma k_e\cdot\begin{pmatrix}\dfrac{b_d\rho_0}{s}+e^{-Ts}\varepsilon(s)+\\ e^{-2Ts}\varepsilon(s)+\cdots+e^{-kTs}\varepsilon(s)\end{pmatrix}=m_d\cdot\delta(s)s^2+b_d\cdot\delta(s)s. \end{aligned} \tag{17}$$

Based on the above equation, the adaptive impedance model proposed in this section represents a typical linear time-invariant system. To establish a transfer function model between the actual contact force error $\varepsilon(s)$ and the predicted contact force error $\delta(s)$, it is necessary to assume that the initial adaptive parameter $\rho_0=0$. The stability conditions associated with (17) can be evaluated through the characteristic equation

$$m_ds^2+b_ds+k_e+k_e\sigma\cdot(e^{-kTs}+\cdots+e^{-Ts})=0. \tag{18}$$

The parameter k is assumed to be a large number, and the denominator in (18) can be rewritten as follows:

$$m_ds^2+b_ds+k_e+k_e\sigma\cdot\frac{e^{-Ts}}{1-e^{-Ts}}=0. \tag{19}$$

The control cycle $T\ll 1$ for lots of robot control systems. Therefore, let $e^{-Ts}\approx 1-Ts$, (19) is rewritten as follows:

$$Tm_ds^3+b_dTs^2+k_eT(1-\sigma)s+k_e\sigma=0. \tag{20}$$

The equation is calculated based on the Routh–Hurwitz criterion as shown in follows:

$$\begin{cases}\dfrac{b_dk_eT^2(1-\sigma)-T\sigma m_dk_e}{b_dT}>0\\ \sigma k_e>0.\end{cases} \tag{21}$$

The equation is rewritten to obtain the stability condition

$$0<\sigma<\frac{b_d\cdot T}{m_d+b_d\cdot T}. \tag{22}$$

B. Model Predictive Adaptive Impedance Control Scheme

The actual force vector obtained by the force/torque sensor are $F_N^S=[f_x^S,f_y^S,f_z^S,m_x^S,m_y^S,m_z^S]^T$, the force signals in the robot based coordinate system $\{R\}$ are obtained through coordinate transformation, as shown in follows:

$$F_N^R=Ad_F({}_S^RT)\cdot F_N^S \tag{23}$$

where $Ad_F({}_S^RT)\in\mathbb{R}^{6\times 6}$ is the adjoint matrix. The symbol F_N^R is the actual force signals in the coordinate $\{R\}$ and $F_N^R=[f_x^R,f_y^R,f_z^R,m_x^R,m_y^R,m_z^R]^{\mathrm{T}}$. Since the x-direction of coordinate $\{R\}$ coincides with the z-direction in coordinate $\{T\}$, so f_x^R is regarded as the actual normal contact force, i.e., $f_x^R=f_N$. Let

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the force state vector be $\tilde{F}_N^R = [f_N, 0, 0, 0, 0, 0]^T$, and the state vector of the MPC scheme is as follows:

$$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} q \\ \dot{q} \\ \tilde{F}_N^R \\ p \end{bmatrix} \tag{24}$$

where $q \in \mathbb{R}^{7\times 1}$ and $\dot{q} \in \mathbb{R}^{7\times 1}$ are. respectively, robot joint position and velocity, $p \in \mathbb{R}^{6\times 1}$ is the robot end pose. The control scheme is established in the discrete-time domain. Based on the differential kinematic models proposed in Siciliano et al. [26], the predictive equation of p in the time t is as follows:

$$p(t+T) = p(t) + T \cdot (J\dot{q}(t) + \Delta\dot{p}(t)) \tag{25}$$

where $\Delta\dot{p} = \dot{p}_d - \dot{p}$, and $\Delta\dot{p}(t) \in \mathbb{R}^{6\times 1}$ is the offset of the robot end velocity, and also is the input for the control scheme at the time t. The symbol $p(t) \in \mathbb{R}^{6\times 1}$ is the current robot end pose. According to (5), the dynamic equation of the state variable $\tilde{F}_N^R$ is as follows:

$$\dot{\tilde{F}}_N^R = K_e(\dot{p}_r - \dot{p}) \tag{26}$$

where $K_e = \text{diag}[k_e, 0, 0, 0, 0, 0]$ and $\dot{p}_r \in \mathbb{R}^{6\times 1}$ represents the robot end's velocity when the contact state is equilibrium. The equation of $\dot{p}_r - \dot{p}$ is the error between the actual robot end velocity and the end velocity in equilibrium. The predictive equation of $\tilde{F}_N^R$ is as follows:

$$\begin{aligned} &\tilde{F}_N^R(t+T) = \tilde{F}_N^R(t) + T \cdot \dot{\tilde{F}}_N^R(t) = F_{Nd} + M_d(\ddot{p}_d(t) - \ddot{p}(t)) \\ &+ B_d(\dot{p}_d(t) - \dot{p}(t) + \tilde{\rho}(t)) + T \cdot K_e \cdot (\dot{p}_r(t) - \dot{p}(t)) \end{aligned} \tag{27}$$

where $F_{Nd} = [f_{Nd}, 0, 0, 0, 0, 0]^T$, $M_d = \text{diag}[m_d, 0, 0, 0, 0, 0]^T$ and $B_d = \text{diag}[b_d, 0, 0, 0, 0, 0]^T$, $\tilde{\rho}(t) = [\rho(t), 0, 0, 0, 0, 0]^T$. The robot end acceleration is usually so small that it can be approximately ignored in this predictive equation. In addition, the components of $\dot{p}_r$ in the force direction are set to zero. Equation (27) is rewritten as

$$\begin{aligned} &\tilde{F}_N^R(t+T) \approx F_{Nd}(t) + B_d \cdot (u(t) + \tilde{\rho}(t)) - T \cdot K_e \cdot \dot{p}(t) \\ &= F_{Nd} + B_d \cdot u(t) + B_d \cdot \tilde{\rho}(t) - T \cdot K_e \cdot J \cdot \dot{q}(t) \end{aligned} \tag{28}$$

where $u(t) = \dot{p}_d(t) - \dot{p}(t)$, i.e., $u(t) = \Delta\dot{p}(t)$, and $J \in \mathbb{R}^{6\times 7}$ is the Jacobian matrix corresponding to the robotic joint position $q(t)$. Substituting (7) into (28)

$$\begin{aligned} &\tilde{F}_N^R(t+T) = F_{Nd}(t) + B_d \cdot u + B_d \cdot \tilde{\rho}(t-T) \\ &+ B_d \cdot \sigma \cdot (F_{Nd}(t-T) - \tilde{F}_N^R(t-T)) - T \cdot K_e \cdot J \cdot \dot{q}(t). \end{aligned} \tag{29}$$

Furthermore, the offset $\Delta\dot{p}(t)$ is approximately argued as the increment of the robot end velocity in the current cycle, so the prediction equation of $\dot{q}(t)$ is

$$\dot{q}(t+T) = \dot{q}(t) + J^+ \cdot u(t) \tag{30}$$

where $J^+ \in \mathbb{R}^{7\times 6}$ is the pseudo inverse matrix of J. Therefore, based on Equations (25), (29), and (30), the state variable $v(t+T)$ in the next control cycle is expressed by the following equation:

$$v(t+T) = \begin{bmatrix} q(t) + T \cdot \dot{q}(t) \\ \dot{q}(t) + J^+ \cdot u(t) \\ F_{Nd}(t) + B_d \cdot \begin{bmatrix} u + \tilde{\rho}(t-T) \\ +\sigma \cdot (F_{Nd}(t-T) \\ -\tilde{F}_N^R(t-T)) \end{bmatrix} - T \cdot K_e \cdot J \cdot \dot{q}(t) \\ p(t) + T \cdot (J\dot{q}(t) + u(t)) \end{bmatrix}. \tag{31}$$

Equation (31) is rewritten as

$$v(t+T) = A \cdot v(t) + D \cdot u(t) + G \tag{32}$$

where A, D, and G are

$$A = \begin{bmatrix} I_{7\times 7} & T \cdot I_{7\times 7} & 0 & 0 \\ 0 & I_{7\times 7} & 0 & 0 \\ 0 & -T \cdot K_e \cdot J & 0 & 0 \\ 0 & T \cdot J & 0 & I_{6\times 6} \end{bmatrix}, D = \begin{bmatrix} 0 \\ J^+ \\ B_d \\ T \cdot I_{6\times 6} \end{bmatrix},$$

$$G = \begin{bmatrix} 0 \\ 0 \\ F_{Nd} + B_d \cdot \begin{pmatrix} \tilde{\rho}(t-T) \\ +\sigma \cdot (F_{Nd} - \tilde{F}_N^R(t-T)) \end{pmatrix} \\ 0 \end{bmatrix}.$$

Furthermore, the cost function is

$$\begin{aligned} \Omega = \min_{\Delta\dot{p}} \sum_{i=0}^{N-1} &\begin{pmatrix} \left(v^{t|iT} - v_d^{t|iT}\right)^T Q \left(v^{t|iT} - v_d^{t|iT}\right) + (\Delta\dot{p}_i)^T R \Delta\dot{p}_i \\ + \left(v^{t|NT} - v_d^{t|NT}\right)^T Q \left(v^{t|NT} - v_d^{t|NT}\right) \end{pmatrix} \\ \text{s.t.} \quad & \alpha_{\min} \le {}^{(J^+(v^{t|iT})\Delta\dot{p}_i)}\Big/_T \le \alpha_{\max} \end{aligned} \tag{33}$$

where $v(t)$ also can be rewritten as v^t. The weight matrix $Q = \text{diag}[Q_q, Q_v, Q_f, Q_p]$, where $Q_q \in \mathbb{R}^{7\times 7}$ and $Q_v \in \mathbb{R}^{7\times 7}$ are, respectively, the weight matrix of the robotic joint position and velocity, $Q_f \in \mathbb{R}^{6\times 6}$ and $Q_p \in \mathbb{R}^{6\times 6}$ are, respectively, the weight matrixes of the interaction force and the robot end pose. The matrix $R \in \mathbb{R}^{6\times 6}$ is the weight matrix for the inputs. $v^{t|iT}$ is the ith predictive state variable in the prediction horizon of the current state variable $v(t)$. $w_{\max} \in \mathbb{R}^{7\times 1}$ and $w_{\min} \in \mathbb{R}^{7\times 1}$ are, respectively, the maximum and minimum of the robotic joint acceleration. N is the prediction horizon length. The ideal state variable $v_d^{t|iT}$ is

$$v_d^{t|iT} = \begin{bmatrix} q_{d,t+iT} \\ \dot{q}_{d,t+iT} \\ F_{Nd} \\ p_{d,t+iT} \end{bmatrix}. \tag{34}$$

According to the calculation process of the ideal moving trajectory of the robot end in the workspace in Section II, the ideal pose matrix of the robot end at time $t+iT$ can be defined as ${}_{P_{t+iT}}^{R}T$. The vector $q_{d,t+iT}$ is calculated based on the matrix ${}_{P_{t+iT}}^{R}T$ through the inverse kinematic model. The joint velocity $\dot{q}_{d,t+iT}$

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is calculated by $(q_{d,t+(i+1)T} - q_{d,t+iT})/T$. The vector $p_{d,t+iT}$ is calculated by transforming the matrix ${}^{R}_{P_{t+iT}}T$. Based on this, (33) can be rewritten as

$$\Omega = \left(\underbrace{\begin{bmatrix} v^t \\ v^{t|T} \\ \vdots \\ v^{t|NT} \end{bmatrix}}_{V_t} - \underbrace{\begin{bmatrix} v_d^t \\ v_d^{t|T} \\ \vdots \\ v_d^{t|NT} \end{bmatrix}}_{V_{dt}} \right)^T \cdot \underbrace{\begin{bmatrix} Q & & \\ & \ddots & \\ & & Q \end{bmatrix}}_{\bar{Q}} \cdot \left(\begin{bmatrix} v^t \\ v^{t|T} \\ \vdots \\ v^{t|NT} \end{bmatrix} - \begin{bmatrix} v_d^t \\ v_d^{t|T} \\ \vdots \\ v_d^{t|NT} \end{bmatrix} \right) + \underbrace{\begin{bmatrix} \Delta\dot{p}^{t|T} \\ \vdots \\ \Delta\dot{p}^{t|NT} \end{bmatrix}^T}_{U_t^T} \cdot \underbrace{\begin{bmatrix} R & & \\ & \ddots & \\ & & R \end{bmatrix}}_{\bar{R}} \cdot \begin{bmatrix} \Delta\dot{p}^{t|T} \\ \vdots \\ \Delta\dot{p}^{t|NT} \end{bmatrix}. \tag{35}$$

According to (32), the vector V_t is calculated by the following equation:

$$V_t = \bar{A}_t \cdot v^t + \bar{D}_t \cdot U_t + \bar{G}_t$$

$$\bar{A}_t = \begin{bmatrix} A \\ (A)^2 \\ \vdots \\ (A)^N \end{bmatrix}, \bar{D}_t = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ D & 0 & \cdots & 0 \\ AD & D & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ (A)^{N-1}D & (A)^{N-2}D & \cdots & AD \end{bmatrix},$$

$$\bar{G}_t = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} A \\ (A)^2 \\ \vdots \\ (A)^N \end{bmatrix} G. \tag{36}$$

Substituting (36) into (35)

$$\begin{aligned} &\Omega = (U_t)^T \cdot H_t \cdot U_t + \Psi_t \cdot U_t + \Theta_t \\ &H_t = \bar{R} + (\bar{D}_t)^T \bar{Q} \bar{D}_t, \Psi_t = (v^t)^T (\bar{A}_t)^T \bar{Q} \bar{D}_t + 2(V_{dt})^T \bar{Q} \bar{D}_t, \\ &\Theta_t = (v^t)^T (\bar{A}_t)^T \bar{Q} \bar{A}_t v^t + 2(v^t)^T (\bar{A}_t)^T \bar{Q} \bar{G}_t + (\bar{G}_t)^T \bar{Q} \bar{G}_t \\ &\quad + (V_{dt})^T \bar{Q} V_{dt} - 2(V_{dt})^T \bar{Q} (\bar{G}_t + \bar{A}_t v^t). \end{aligned} \tag{37}$$

Equation (37) is regarded as a typical quadratic programming problem and the optimal solution $U_t^* = [\Delta\dot{p}_i^{t|T*}, \Delta\dot{p}_i^{t|2T*}, \ldots, \Delta\dot{p}_i^{t|NT*}]$ is obtained by solving the problem, where $\Delta\dot{p}_i^{t|T*}$ is the control input of the controller in the control cycle and used to compensate for the expected robot end pose in the next cycle.

C. Moving Parameters Optimization

The controllers with fixed parameters are difficult to ensure the control performance of the contact force under time-varying contact environment. Therefore, the damping parameter b_d and the contact environment stiffness k_e, involved in the impedance model, are adjusted in real time to further enhance the control performance of the controller. Equation (29) is rewritten as

$$\begin{cases} \tilde{F}_N^R(t+T) \approx F_{Nd}(t) + \Lambda(t) \cdot \eta(t) \\ \Lambda(t) = \left[u(t) + \tilde{\rho}(t-T) + \sigma\left[F_{Nd}(t-T) - F_N^R(t-T) \right], -TJ\dot{q}(t) \right] \\ \eta(t) = \begin{pmatrix} B_d(t) \\ K_e(t) \end{pmatrix}. \end{cases} \tag{38}$$

Let $\eta(t)$ be the input of the whole moving parameters optimization process at time t, and the corresponding cost function is

$$J = \sum_{i=t-(N_b-1)T}^{t} \left[\bar{F}_N^R(i) - \tilde{F}_N^R(i) \right]^T \left[\bar{F}_N^R(i) - \tilde{F}_N^R(i) \right] \tag{39}$$

where N_b is the moving horizon length, $\bar{F}_N^R = I_F \cdot F_N^R$ and $I_F = \mathrm{diag}[1, 0, 0, 0, 0, 0]$, and $\tilde{F}_N^R(i)$ is the actual normal contact force in the ith control cycle. The constraints are

$$-\tau_{\max} < J(q(t))^T \cdot [F_{Nd}(t) + \Lambda(t)\eta(t)] < \tau_{\max} \tag{40}$$

where $\tau_{\max} \in \mathbb{R}^{7\times 1}$ is the maximum torque for the robot joints. Equation (38) is rewritten as

$$J = \sum_{i=t-(N_b-1)T}^{t} \bar{F}_N^R(i)^T \cdot \bar{F}_N^R(i) - 2\bar{F}_N^R(i)^T \cdot \tilde{F}_N^R(i) + \tilde{F}_N^R(i)^T \cdot \tilde{F}_N^R(i). \tag{41}$$

Substituting (38) into (41)

$$\begin{aligned} J &= \sum_{i=t-(N_b-1)T}^{t} \begin{pmatrix} {F_{Nd}}^2 + 2F_{Nd}\Lambda(i-T)\eta + (\Lambda(i-T)\eta)^2 \\ -2\bar{F}_N^R(i-T)\Lambda(i-T)\eta \end{pmatrix} \\ &= \underbrace{\sum_{i=t-(N_b-1)T}^{t} {F_{Nd}}^2}_{M_t} + 2 \underbrace{\sum_{i=k-N+1}^{k} (F_{Nd} - \bar{F}_N^R(i-T))\Lambda(i-T)}_{N_t} \cdot \eta \\ &\quad + \eta^T \underbrace{\sum_{i=t-(N_b-1)T}^{t} \Lambda(i-T)^T \Lambda(i-T)}_{C_t} \cdot \eta = 2N_t\eta + \eta^T C_t \eta + M_t. \end{aligned} \tag{42}$$

The matrix M_t is the constant matrix, so (42) is rewritten as follows:

$$\begin{cases} J_{QP} = 2N_t\eta + \eta^T C_t \eta \\ -\tau_{\max} < J^T \cdot F_{Nd} + J^T \Lambda(t)\eta < \tau_{\max}. \end{cases} \tag{43}$$

Equation (43) is solved as a typical quadratic programming problem, and the corresponding optimal solution $\eta^* = [B_d^*, K_e^*]^T$ in the current control cycle is obtained by solving the problem. Furthermore, the parameters B_d^* and K_e^* are used in the force prediction in the next control cycle. The whole control scheme of the proposed controller MPAIC-MPO is shown in Fig. 4.

IV. Simulation

The simulation platform is constructed by using MATLAB 2019b and CoppeliaSim 4.0.0, as expressed in Fig. 5. The moving path of the robot end in this platform is determined in

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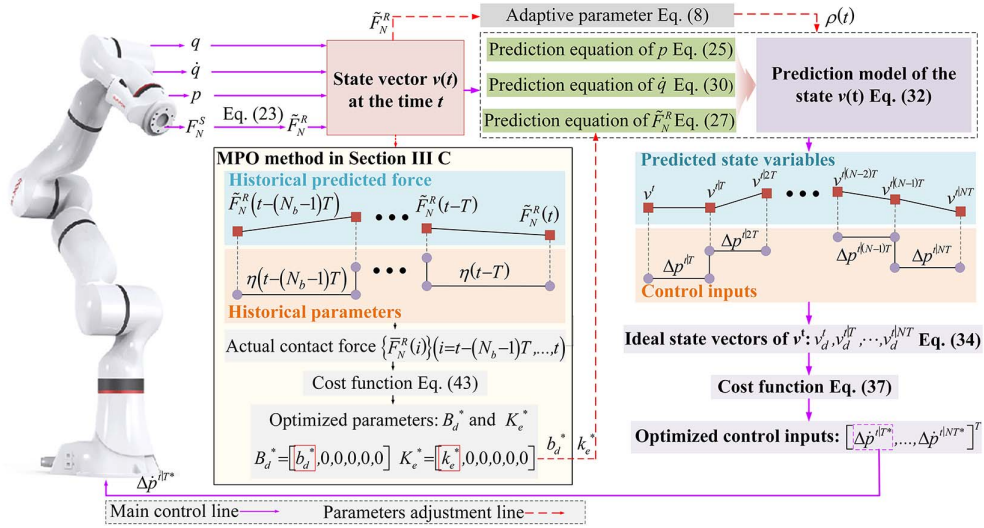


Fig. 4. Control scheme of the controller MPAIC-MPO.

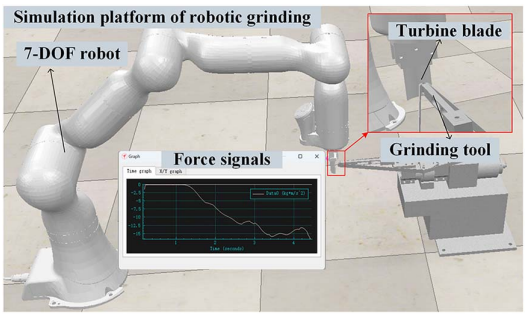


Fig. 5. Simulation platform of the robotic grinding system.

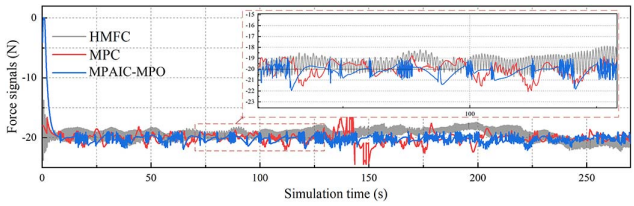


Fig. 6. Control performance in normal contact force.

TABLE II
PARAMETERS IN MPAIC-MPO CONTROLLER

Symbol	Parameter Name	Value
K_e	Stiffness coefficient matrix	diag(10 000, 0, 0, 0, 0, 0)
B_d	Damping coefficient matrix	diag(200, 0, 0, 0, 0, 0)
M_d	Inertial coefficient matrix	diag(1, 0, 0, 0, 0, 0)
Q_P	Position weight matrix	diag(0, 1000, 1000, 10, 10, 10)
Q_F	Force weight matrix	diag(1000, 0, 0, 0, 0, 0)
R	Input weight matrix	diag(10, 10, 10, 10, 10, 10)
N	Prediction horizon length	50
N_b	Moving horizon length	80
Σ	Adaptive rate	0.11

Section II. The ideal normal contact force is 20 N during the moving process of the robot end-effector along the planned trajectory distributed on the blade surface. Table II shows the key control parameters involved in the proposed controller through parameter iteration tests. The inertial parameter m_d is set to 1 based on the Chen et al. [27]. The damping parameter b_d mainly influences the overshoot and steady-state errors during the contact force control. The stiffness parameter k_e is the main factor for the fluctuation of the normal contact force. The parameters k_e and b_d are, respectively, set to 10 000 N/m and 200 based on the calculation method of nominal damping in Chen et al. [27]. In Section III-A, the range of the parameter σ is calculated based on (22). The adaptive rate can be calculated as follows:

$$\sigma = \kappa \cdot \frac{b_d \cdot T}{m_d + \mathrm{b}_d \cdot T} \quad (0 < \kappa < 1) \tag{44}$$

where the coefficient $\kappa = 0.65$ based on parameter iteration tests. The control cycle T in the simulation is 0.001 s, and the range is $0 < \sigma < 0.17$, so the parameter σ is set to 0.11. If the prediction horizon length and moving horizon length are too long, it will lead to a longer control cycle, which is not conducive to real-time control. Therefore, the prediction horizon length N and the moving horizon length N_b are set to 50 and 80 by iteration tests. The weight matrices corresponding to the robotic joints' position and velocity are set to zero matrices.

To validate the effectiveness of the developed controller, the typical hybrid motion/force controller (HMFC) [26] and the typical MPC algorithm [20] are used as comparison controllers. The force control accuracy is calculated by the following equation:

$$e_F = \left(\frac{\sum_{i=1}^{N_{\text{sample}}} (f_{Ni} - f_{Nd})^2}{N_{\text{sample}}} \right)^{0.5} \tag{45}$$

where N_{sample} is the total of the sampling points and f_{Ni} is the ith actual normal contact force. Fig. 6 shows the force signals with different controllers. The force tracking accuracy is 1.717 N (HMFC), 1.085 N (MPC), and 0.886 N (MPAIC-MPO).

For the robot end pose $p = [x, y, z, w, \beta, r]^T$, p_x is located in the force direction (x-direction of the coordinate $\{R\}$) and is not considered in the control of the robot end motion. Therefore, let

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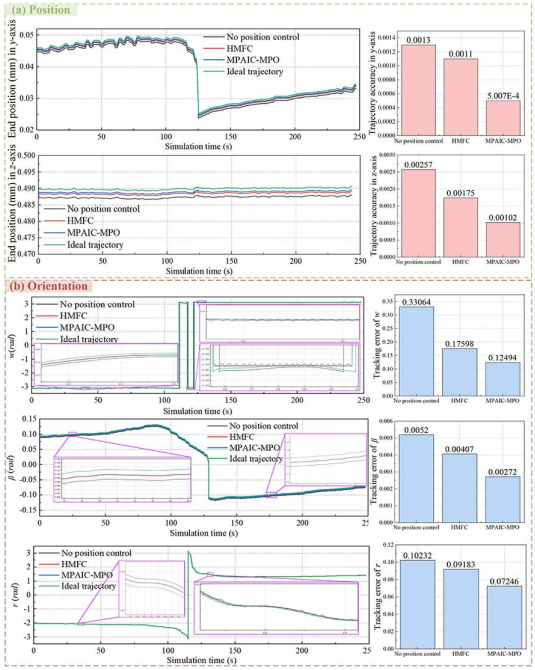


Fig. 7. Trajectory tracking of the robot end in the simulation process.

TABLE III
POSITION/ORIENTATION CONTROL ACCURACY

Controllers	No Position Control	HMFC	MPAIC-MPO
Position control accuracy	0.0019 mm	0.0014 mm	0.0011 mm
Orientation control accuracy	0.1261 rad	0.0906 rad	0.0667 rad

the vectors $p_p = [y, z]^T$ and $o = [w, \beta, r]^T$ be the position vector and orientation vector. The vector o is represented by Euler angles. In addition, the ideal robot end poses $p_d = [x_d, p_{pd}, o_d]^T$. The position control accuracy e_p and posture control accuracy e_o of the robot end is calculated by the following equation:

$$e_P = \frac{1}{N_{\text{path}}} \cdot \sum_{i=1}^{N_{\text{path}}} (\|p_{pi} - p_{\text{pdi}}\|_2), \quad e_o = \frac{1}{N_{\text{path}}} \cdot \sum_{i=1}^{N_{\text{path}}} (\|o_i - o_{di}\|_2) \tag{46}$$

where N_{path} is the total of the sampling points distributed at the path tracking. The calculation theory of the robotic end pose control in Cartesian space using the proposed controller is the same as the position control theory involved in the typical MPC algorithm. Therefore, HMFC and no position control are used as two comparison methods to verify the path tracking of the robot end using the proposed controller. Fig. 7 shows the trajectory of the robot in Cartesian space, and control accuracy of position and orientation are expressed in Table III.

The position control accuracy with MPAIC-MPO is improved by 41.96% and 21.19% compared to that with no position control and HMFC. In addition, the orientation control accuracy with MPAIC-MPO is improved by 47.11% and 26.38% compared to that with no position control and HMFC.

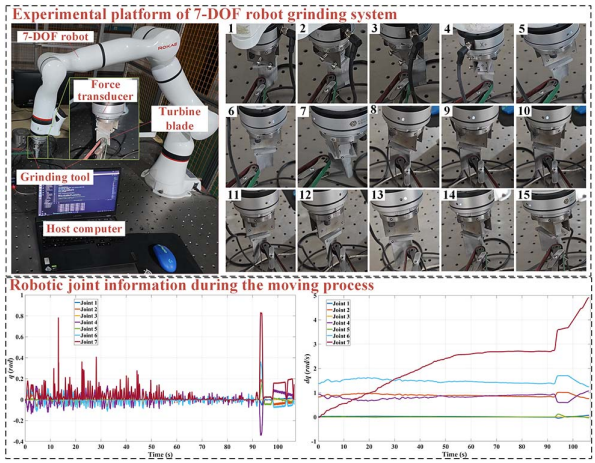


Fig. 8. Experimental platform of the robotic grinding system.

TABLE IV
DETAILED INFORMATION OF KWR75B

F_x, F_y, F_z	M_x, M_y, M_z	Resolution	Precision	Sampling Frequency
200 N	8 Nm	0.03 %FS	0.1 %FS	1000 HZ

TABLE V
CRUCIAL PARAMETERS OF THE CONTROLLER MPAIC-MPO

Symbol	Parameter Name	Value
K_e	Stiffness coefficient matrix	diag(35981, 0, 0, 0, 0, 0)
B_d	Damping coefficient matrix	diag(379, 0, 0, 0, 0, 0)
M_d	Inertial coefficient matrix	diag(1, 0, 0, 0, 0, 0)
Σ	Adaptive parameter	0.18
N	Prediction horizon length	15
N_b	Moving horizon length	30

V. EXPERIMENT

Fig. 8 shows the actual robot grinding platform. The seven-DOF industrial ROAKE xMate7 Pro is used in the validation. In addition, the robot feed rate v_w is about 8 mm/s. The maximum control frequency is 1000 Hz. The force transducer used in this experiment is KWR75B, and its information is expressed in Table IV. The ideal trajectory distributed in the turbine blade surface is also shown in Fig. 2 in Section II, and the corresponding moving trajectory of the end is also L. The ideal normal contact force is set to 4 N.

The crucial control parameters involved in the controller are illustrated in Table V. The inertial parameter m_d is set to 1. The initial values of parameters k_e and b_d are, respectively, set to 35981 N/m and 379 N/m through the multiple repeated tests of the environmental stiffness and the calculation method of nominal damping [27]. Furthermore, the value range of the adaptive parameter σ is $0 < \sigma \leq 0.27$. The parameter σ is set to 0.18 based on (44). The parameters N and N_b are set to 15 and 30 by iteration tests like the tests in the simulation. In addition, the weight matrices Q_q, Q_v, Q_f, and Q_p are all the same as those used in Section IV.

MPO is used to real-time modify the environmental stiffness and the damping coefficient to improve the predictive accuracy

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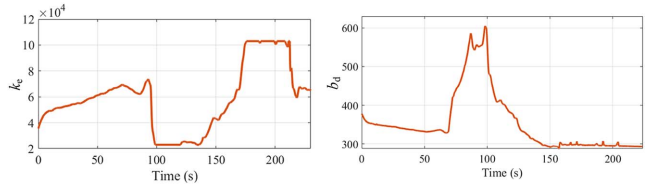


Fig. 9. Changing in b_d and k_e.

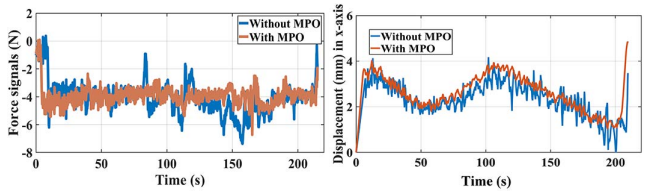


Fig. 10. Comparison in the force signals with and without MPO.

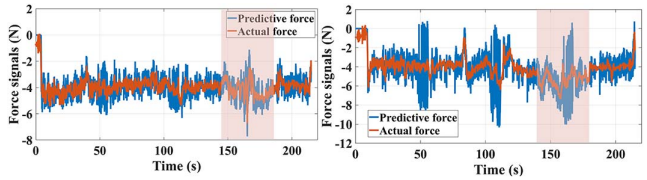


Fig. 11. Predictive force signals with and without MPO.

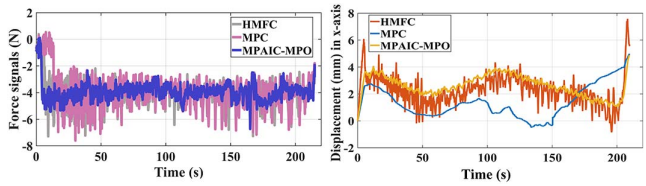


Fig. 12. Comparison of the force signals with different controllers.

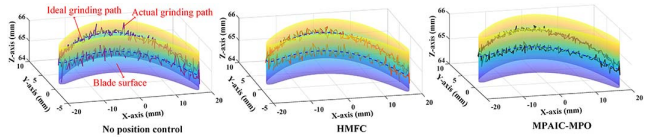


Fig. 13. Actual moving path of the grinding tool relative to the turbine blade.

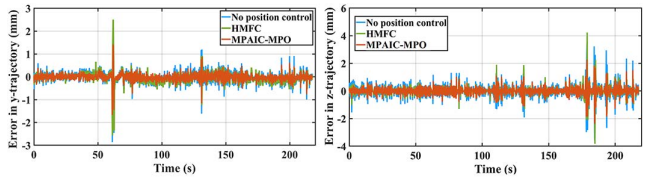


Fig. 14. Position errors of the robot end in y- and z-directions.

of the actual contact force. Fig. 9 shows the modification of the parameters b_d and k_e in every control cycle.

To verify that MPO can effectively improve the prediction accuracy of normal contact force in the model prediction process, the corresponding experiment has been conducted. The feed speed, ideal normal grinding force, robot end moving path, and key control parameters remain unchanged. Fig. 10 shows the comparison in the actual normal contact force with and without MPO during the moving process. As shown in this figure, the tracking accuracy of the force signals with MPO is better than that without MPO, especially in the stage of 140–180 s. In addition, the compensation displacement in the force direction is also expressed in Fig. 10. The compensation displacement changes of the robot end pose in the force direction using MPO is significantly smoother than that without MPO. The control accuracy with and without MPO is 1.834 and 0.744 N, respectively, which is calculated by (45). The signals of the predictive normal contact force with and without MPO are shown in Fig. 11. The MPO method can adjust online the key control parameters k_e and b_d involved in the adaptive impedance model based on the predictive errors of normal contact force during the previous control cycles. The adjusted control parameters can decrease the errors of the impedance model to enhance the predictive accuracy of the normal contact force, thereby improving the force control accuracy. The corresponding predictive accuracy is calculated by the following equation:

$$e_{\text{Pre}} = \left(\frac{1}{N_{\text{sample}}} \sum_{i=1}^{N_{\text{sample}}} \left(f_{N\text{Pre},i} - f_{Ni} \right)^2 \right)^{0.5} \tag{47}$$

where $f_{N\text{Pre},i}$ is the predictive normal contact force. The predictive accuracy with and without MPO is 1.032 and 1.843 N based on this figure. The method MPO can improve the predictive accuracy of the normal contact force.

The comparison experiment for the three controllers (HMFC, MPC, and MPAIC-MPO) is conducted. In the experiment, the three controllers are, respectively, used in the experiment. Meanwhile, the initial robot state, ideal normal contact force, robot end feed rate, and external contact environment remain unchanged. Fig. 12 shows the normal contact force with different controllers in the motion process. In addition, Fig. 12 presents the displacement of the robotic end in the force direction. As shown in the figure, the displacement change with the proposed controller is smoother than that with HMFC. The force control accuracy with the three controllers is 1.276 N (HMFC), 1.213 N (MPC), and 0.744 N (MPAIC-MPO). The adaptive impedance model and MPO method involved in the proposed controller can predict precisely the normal contact force in next control cycles. It can further improve the control accuracy of the normal contact force.

Due to the basic theory of position control in MPC is similar to that in MPAIC-MPO, so the robot motion trajectory with MPC cannot be used in comparison in the field of position control. Fig. 13 shows the actual moving path of the grinding device relative to the turbine blade, and the dashed curve represents the ideal trajectory in the blade surface. As shown in this figure, the fluctuation of the moving path using MPAIC-MPO is significantly smaller than that using HMFC and no position control. Therefore, the proposed controller enhances the path control of the robot compared to the other two methods. Furthermore, the path tracking errors of robot TCP in the y- and z-directions of the coordinate $\{R\}$ are shown in Fig. 14. Based on this, the path control accuracy with the three control methods is calculated by (46).

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TABLE VI
POSITION/ORIENTATION CONTROL ACCURACY

Controllers	No Position Control	HMFC	MPAIC-MPO
Position control accuracy	0.6694 mm	0.5651 mm	0.4468 mm
Orientation control accuracy	0.0089 rad	0.0018 rad	0.0012 rad

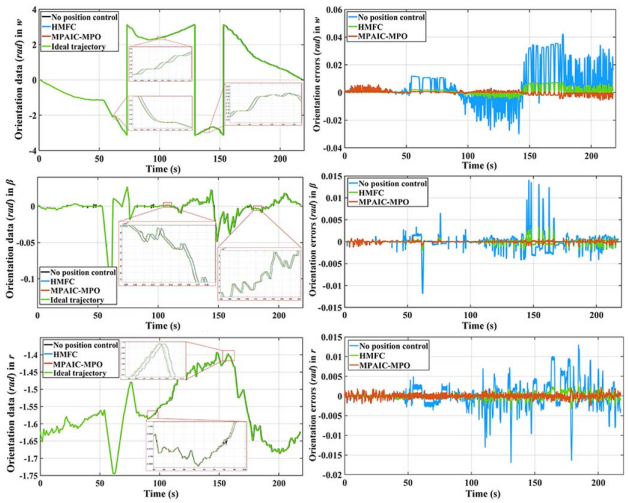


Fig. 15. Actual orientation and errors of the robot end.

Table VI shows the path control accuracy with these different control methods. Furthermore, the actual orientation data of the robot end pose in Euler angles w, β, and r is shown in Fig. 15, moreover, the orientation tracking errors with the use of the three control methods are also shown in Fig. 15. Based on this, the orientation control accuracy with using these methods is calculated by (46) and the statistical results are also shown in Table VI. In the position/attitude control of MPAIC-MPO, the current control input is further calculated by minimizing the errors between the predicted pose and the theoretical pose in the next control cycles. So, it can ensure the minimization of pose errors of robot end during the future several control cycles.

Reproducibility experiments and the test after long-term operation of the experiment conducted in the section are conducted, and the results are consistent with the original experimental results in this section. Furthermore, to further verify that the proposed controller can effectively improve the position/force control accuracy of the robot end, we conduct the similar experiment with a flat plate. The experimental results indicate that the force control accuracy with MPAIC-MPO is improved by 50.04% compared to HMFC. In addition, the position control accuracy with MPAIC-MPO is improved by 48.10% compared to HMFC.

Industrial robots can be used as carriers to verify the control superiority of the proposed controller under the condition of heavy loads in future research. In addition, the robustness of the proposed controller under disturbances such as noise and high temperature will be conducted.

VI. CONCLUSION

1) The adaptive impedance model and the robot kinematic model are developed to construct the MPC scheme. Based on the control scheme, the MPO method is used to adjust adaptively the parameters b_d and k_e based on historical force errors.
2) The force error of the MPAIC-MPO controller is improved by 48.39% and 18.34% compared to that with HMFC and MPC in the simulation. The position control accuracy with MPAIC-MPO is improved by 41.96% and 21.19% compared to that with no position control and HMFC.
3) Experimental validation of the robotic force/motion control is conducted. The force error of the MPAIC-MPO controller is improved by 44.69% and 38.66% compared to that with HMFC and MPC. The position control accuracy with MPAIC-MPO is enhanced by 33.25% and 21.34% compared to that with no position control and HMFC.

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